RouteLLM BERT on RouterBench: Results and Why We Chose RouteLLM

Draft for team review · Prepared 5 November 2026 · Based on the Colab notebook (rpcnb.pdf) and the team's planning documents

1. Summary

We tested the pretrained RouteLLM BERT router against random routing on 500 RouterBench prompts (300 GSM8K, 200 ARC-Challenge), using RouterBench's stored GPT-4 and Mixtral answers and costs, so no API calls were needed.

- The router did **not** beat random routing. Accuracy gains over random were between −0.010 and +0.006 at cut-offs from 10% to 90%.
- At matching accuracy the router cost **more** than random routing at every cut-off, and the 95% interval excludes zero from 30% to 90%.
- The router mostly separated the two datasets (GSM8K versus ARC) and showed no clear skill at telling easy from hard prompts within a dataset.
- This is consistent with our claim that RouteLLM is a **suitable and representative** router for our study, not the best one. We report it as it is.

2. Experiment setup (as run in the notebook)

Item	Choice
Data	RouterBench 0-shot file (36,497 rows). Kept GSM8K ("grade-school-math", 7,450 prompts) and ARC-Challenge (1,470 prompts): 8,920 prompts after dropping missing values
Strong model	gpt-4-1106-preview (stored score and cost)
Weak model	mistralai/mixtral-8x7b-chat (stored score and cost). Our earlier router setup names Mixtral-8x7B-Instruct-v0.1, so the two are related but not identical
Router	RouteLLM BERT (routellm/bert_gpt4_augmented), scoring only the extracted question text

Item	Choice
Compute	CPU only (no GPU). About 2.6 to 2.8 prompts per second, so the full 8,920 prompts would take about 55 minutes
Sample	Fixed-seed random sample (seed 42): 300 GSM8K + 200 ARC = 500 prompts
Routing rule	Send the top share of prompts, ranked by BERT score, to GPT-4; the rest go to Mixtral. Shares 10% to 90% in steps of 10
Cut-off scores	Set on our own prompts (the score of the prompt at each share), not reused from the Arena-based cut-offs
Baselines	Random routing (expected value at the same share), always-strong, always-weak, and an oracle (cheapest model that is correct)
Metrics	Mean score, total cost, accuracy gain over random, and cost saved at matching accuracy
Uncertainty	Bootstrap, 2,000 resamples, 95% percentile intervals

Cleaning note: the stored prompts are lists that include few-shot examples, so the notebook parses the list and takes the last element as the question. After this fix all 8,920 questions are unique.

Cost saved at matching accuracy is defined as follows: find the random strong/weak mix that reaches the router's accuracy, take its cost, and subtract the router's cost. A negative value means the router costs more.

3. Results

3.1 Dataset properties in RouterBench (full 8,920 prompts)

Dataset	Prompts	GPT-4 accuracy	Mixtral accuracy	GPT-4 total cost	Mixtral total cost
GSM8K	7,450	0.659	0.519	63.68	1.97

Dataset	Prompts	GPT-4 accuracy	Mixtral accuracy	GPT-4 total cost	Mixtral total cost
ARC-Challenge	1,470	0.962	0.832	1.35	0.08

GPT-4 adds about the same accuracy on both (about +0.14 on GSM8K, +0.13 on ARC), but a GPT-4 call on GSM8K costs about $0.0086 per prompt against about $0.0009 on ARC, roughly nine times more. GSM8K scores are stored in steps of 0.25; ARC scores are 0 or 1.

3.2 BERT scores on the 500-prompt sample

Dataset	Prompts	Mean BERT score	Min	Max
ARC-Challenge	200	0.397	0.144	0.833
GSM8K	300	0.525	0.270	0.807

BERT scores GSM8K higher on average, so ranking by score sends GSM8K prompts to GPT-4 first.

3.3 Baselines (500 prompts)

Method	Mean score	Total cost (USD, RouterBench 2024 prices)
Always strong (GPT-4)	0.7735	2.6961
Always weak (Mixtral)	0.6350	0.0912
Oracle (cheapest correct)	0.7940	1.1042

The oracle shows that good routing could in principle match GPT-4 at well under half its cost. This router does not capture that.

3.4 Router versus random: accuracy

Target share to GPT-4	Cut-off score	Actual share (all)	Share of GSM8K to GPT-4	Share of ARC to GPT-4	Router accuracy	Random accuracy	Gain over random	95% CI lower bound
10%	0.6256	0.10	0.120	0.070	0.6460	0.6488	−0.0029	−0.0116
20%	0.5734	0.20	0.257	0.115	0.6605	0.6627	−0.0022	−0.0126
30%	0.5346	0.30	0.413	0.130	0.6725	0.6766	−0.0040	−0.0174

Target share to GPT-4	Cut-off score	Actual share (all)	Share of GSM8K to GPT-4	Share of ARC to GPT-4	Router accuracy	Random accuracy	Gain over random	95% CI lower bound
40%	0.5065	0.40	0.573	0.140	0.6840	0.6904	−0.0064	−0.0202
50%	0.4740	0.50	0.717	0.175	0.6995	0.7042	−0.0048	−0.0197
60%	0.4492	0.60	0.853	0.220	0.7085	0.7181	−0.0096	−0.0227
70%	0.4058	0.70	0.937	0.345	0.7340	0.7320	+0.0020	−0.0093
80%	0.3652	0.80	0.980	0.530	0.7520	0.7458	+0.0062	−0.0065
90%	0.3148	0.90	0.993	0.760	0.7615	0.7596	+0.0019	−0.0098

The upper confidence bounds were cut off in the PDF, so confirm them in results_router_vs_baselines.csv before stating that every interval includes zero. Every lower bound is below zero.

3.5 Router versus random: cost at matching accuracy

Target share	Router cost	Random cost (same share)	Cost saved at matching accuracy	% saved	95% CI
10%	0.4241	0.3517	−0.1260	−42.3%	−0.2990 to +0.1153
20%	0.7556	0.6122	−0.1848	−32.4%	−0.3972 to +0.0706
30%	1.1529	0.8727	−0.3564	−44.7%	−0.6242 to −0.1053
40%	1.5485	1.1331	−0.5357	−52.9%	−0.8139 to −0.2438
50%	1.9111	1.3936	−0.6068	−46.5%	−0.8905 to −0.3407
60%	2.2266	1.6541	−0.7530	−51.1%	−1.0025 to −0.4523
70%	2.4391	1.9146	−0.4860	−24.9%	−0.7109 to −0.1853
80%	2.5734	2.1751	−0.2817	−12.3%	−0.5173 to −0.0882
90%	2.6445	2.4356	−0.1742	−7.1%	−0.3824 to −0.0242

Negative means the router costs more than a random mix at the same accuracy. At the 50% cut-off the router costs about 47% more.

3.6 Within each dataset (cut-offs 20% and 50% only; the 80% output was cut off)

Dataset	Prompts	20% cut-off	50% cut-off
GSM8K	300	router 0.557 vs random 0.558 (−0.002)	router 0.595 vs random 0.599 (−0.004)
ARC-Challenge	200	router 0.820 vs random 0.819 (+0.001)	router 0.880 vs random 0.863 (+0.017)

GSM8K is flat. ARC is flat at 20% and slightly positive at 50%, but this is one cut-off on 200 prompts with no confidence interval, so it should not be called a real effect.

3.7 Charts

Three charts were saved in the MLResearch folder: accuracy versus cost, cost saved at matching accuracy, and who gets sent to the strong model. In the PDF, Chart 1 is cut off and Chart 3 is not visible, so open the PNGs and check them. Label Chart 2's axis "cost saved vs random (negative = router costs more)".

4. Interpretation (our reading, not a tested claim)

- The router mostly sorts prompts by dataset, not by difficulty. At the 50% target it sends 71.7% of GSM8K prompts to GPT-4 but only 17.5% of ARC prompts.
- GPT-4 buys a similar accuracy gain on both datasets, but costs about nine times more per prompt on GSM8K. So the router spends the GPT-4 budget where each unit of accuracy is most expensive, which produces the negative saving.
- RouteLLM predicts whether GPT-4 would win and ignores price, so a cost-at-matching-accuracy metric is a harsh test for it. This is a mismatch with the metric, not necessarily a bug. The baselines (0.77, 0.64, 0.79) look sensible, so the pipeline is probably sound, but the code is worth one more check (sort direction, score lookup, cost sums).
- Label these points as interpretation in the paper.

5. Draft results text for the paper

> On a random sample of 500 RouterBench prompts (300 GSM8K, 200 ARC-Challenge; seed 42), the BERT router did not outperform random routing. Accuracy gains over random were between −0.010 and +0.006 across cut-offs from 10% to 90%, with all 95% bootstrap intervals including zero. At matching accuracy the router cost more than random routing at every cut-off (for example 47% more at the 50% cut-off). The router sent most GSM8K prompts to GPT-4 but few ARC prompts, so its choices mostly followed the dataset rather than prompt difficulty, and GPT-4 calls on GSM8K cost about nine times more per prompt. This agrees with the two benchmarks' finding that router differences are small, and it supports our claim that RouteLLM is a suitable, representative router for our study, not the best one.

Adjust the text once the full 8,920-prompt run is finished.

6. Limitations to state

- Random sample of 500 prompts, because no GPU was available. Confidence intervals are wide.
- The router scored only the question, while stored costs cover the full prompt including the few-shot template.
- GSM8K scores are stored in 0.25 steps.
- Cut-offs were set on these prompts, not on RouteLLM's Arena reference data.
- RouterBench costs come from its 2024 data collection, not today's prices.
- One strong and one weak model, English only, and only two exact-match subsets.
- RouterBench's weak model is mixtral-8x7b-chat; our router was set up with Mixtral-8x7B-Instruct-v0.1.
- Only the BERT router was run, so it is not compared with the other four routers.
- The overlap between the BERT checkpoint's training data and GSM8K and ARC is still unchecked. Verify what the RouteLLM paper says about its training data before interpreting the results.

7. Why we chose RouteLLM

7.1 Simple reasons tied to what we did

1. It gives a score, not just a yes/no: a number from 0 to 1 (the estimated chance GPT-4 beats the weak model). The Hinglish study needs this for spread, slope, crossover and cut-off analysis.
2. It makes a strong-versus-weak decision, which maps onto the question of whether Hinglish prompts are sent to the strong model more or less often than their English twins.
3. It is free to run. The BERT router is pretrained, and in our notebook it scored prompts with a placeholder API key and made no model calls.
4. It is open source (Apache-2.0), so others can reproduce the work.
5. It is widely cited, so findings about it matter beyond our project.
6. Its pretrained routers were built for the gpt-4-1106-preview and Mixtral pair, and RouterBench stores results for a GPT-4 and Mixtral pair, so testing is free.
7. It is trained on general human-preference data, not tuned to one benchmark.

7.2 Compared with the other four routers

Router	Why it fits less well for us
Hybrid LLM	Also a strong-versus-weak router, but it needs training and we found no official repository
FrugalGPT	A cascade that generates answers before deciding, so there is no score to read beforehand; needs a trained scorer
GraphRouter	Chooses among many models, so it does not match our binary question; needs training (MIT licence)
EmbedLLM	Chooses among many models; needs training; no stated licence, so call its code "publicly available"

RouteLLM is the only one of the five with pretrained checkpoints, which matters because we have no budget and no GPU.

7.3 Where the others look better (say this ourselves)

- On RouterArena, GraphRouter has higher accuracy (57.0% versus 47.0%) and a better accuracy-cost rank (7 versus 9 in Table 2).
- On LLMRouterBench, GraphRouter's performance-first setting is slightly more accurate (+2.8% versus +2.6%), but it costs about 20% more than GPT-5 alone, while RouteLLM saves about 11%.
- RouteLLM leads RouterArena on cost and robustness, but a binary router has only two outputs, so part of its 100% robustness may be structural.
- Both benchmarks find router differences are small and depend on the model pool and dataset.

8. What to say to the teacher

Question: "How is your model the best compared with other routers in the research papers?"

> We didn't choose RouteLLM because it is the best router. The published benchmarks don't show that, and we don't claim it. We chose it because it is the only one of the five we could run for free, it makes the strong-versus-weak decision our study needs, and it gives a score we can analyse. The benchmarks show no router is clearly best, and our own test found it no better than random routing on GSM8K and ARC.

Why this is safe to show:

- The plan was fixed before running, and the claim was always "suitable and representative, not the best".
- The teacher asked us to run our own experiments. A clean, honestly reported experiment with an unexpected result shows we did.
- The Hinglish study is the main contribution; this experiment only supports the router choice.
- Do not change the data, cut-offs or metric to improve the numbers, and do not leave the result out.
- Frame it as a finding: the router predicts whether GPT-4 would win and ignores price, so a cost-based metric is a harsh test, and the checkpoint was not tuned for these datasets.

9. RouterArena numbers: what to quote

The live leaderboard (opened 5 November 2026) matches Table 6 and Figure 1 of the paper: RouteLLM arena score 48.1, accuracy 47.0%, $0.27 per 1K queries. Table 2 and Appendix H give a second set (rank 9, accuracy 62.24%), and the paper does not explain the difference, so always name the table.

Message for the team:

> We quote accuracy 47.0% and cost $0.27 from Table 6 (RouterArena's figures for RouteLLM with GPT-4o and Mixtral, not for our setup), written as rank 9 in Table 2, rank 11 in Figure 1. OK?

Our inference, not the paper's: 47.0% is almost identical to Mixtral's own 47.01% in Table 5, which suggests most queries went to Mixtral. This explanation does not fit the 62.24% set, so label it as inference.

Do not drop the weak accuracy, because that would look like cherry-picking. Suggested sentence:

> On RouterArena, RouteLLM ranks third on the paper's six-metric average and first on optimal selection, optimal cost and robustness, but only tenth on optimal accuracy and ninth on the accuracy-cost score (Table 2; eleventh in Figure 1).

10. Next steps

1. Send the teacher message if it has not gone out: "Sir, we can't pay for API calls. May we use RouterBench's released GPT-4 and Mixtral answers and costs? Does testing BERT against random, always-strong and always-weak baselines count as running our own experiments, or must other routers be run too? Should the router comparison go in the main paper or an appendix?" If he says no to the first question, the model pair must change.
2. Check the three charts in the MLResearch folder and fix Chart 2's axis label.
3. Rerun on all 8,920 prompts (about 55 minutes on CPU, using the partial-save cell) and add a per-dataset table with confidence intervals.
4. Check the RouteLLM paper for training-data overlap with GSM8K and ARC.
5. Confirm the upper confidence bounds and the 80% within-dataset output in the saved CSVs.

6. Give Person 4 results_router_vs_baselines.csv, results_baselines.csv, the charts, and this document's results text and limitations.
7. Person 4 applies the paper fixes already listed (Section 3.3 file names, English AUC 0.611, Table 6 κ 0.782, gradient-boosting figure about 0.100, new limitations, API-key wording).
8. Optional: the typo and paraphrase robustness check.